# CAPSTONE PROJECT REPORT

# AUTOMATION TESTING FRAMEWORK FOR TEST AUTOMATION PRACTICE WEBSITE


**Submitted By**

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WIPRO TALENTNEXT PROGRAM

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# INDEX

**Project Title:**

Automation Testing Framework for Test Automation Practice Website Using Selenium, TestNG, Maven, Jenkins, and GitHub

**Project Overview:**

This project focuses on automating the Test Automation Practice Website using Selenium WebDriver with Java. The framework is designed using the Page Object Model (POM) design pattern and integrates TestNG, Maven, Jenkins, Apache POI, and GitHub for efficient test automation, reporting, and continuous integration.

The framework automates major web functionalities including form validation, radio buttons, checkboxes, dropdowns, alerts, file uploads, window handling, tables, broken link verification, mouse actions, and date picker validation.

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**Project Objectives:**

- Automate repetitive manual test cases.
- Improve test coverage and execution speed.
- Implement a reusable Page Object Model framework.
- Integrate TestNG for test management.
- Read test data using Apache POI.
- Generate execution reports.
- Integrate GitHub for version control.
- Integrate Jenkins for Continuous Integration.
- Reduce manual testing effort and improve quality.

**Technologies Used:**

| Tool | Purpose |
|---|---|
| Java | Programming Language |
| Selenium WebDriver | Web Automation |
| TestNG | Test Execution Framework |
| Maven | Build Management |
| Apache POI | Excel Data Handling |
| Jenkins | Continuous Integration |
| GitHub | Source Code Management |
| Extent Reports | Reporting |
| Chrome Browser | Test Execution |

**Project Website:**

https://testautomationpractice.blogspot.com/

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**Framework Design:**

The framework follows the Page Object Model (POM) design pattern.

**Project Structure:**

**CapstoneProject**

```
├── src/test/java
|   ├── base
|   ├── pages
|   ├── testcases
|   ├── listeners
|   └── utilities
└── src/test/resources
    ├── pom.xml
    ├── testng.xml
    ├── Jenkinsfile
```

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**Features Implemented :**

**Selenium WebDriver**

- Browser Launch
- Element Identification
- Form Handling
- Validation

**TestNG**

- Test Prioritization
- Assertions
- Test Execution
- Reporting

**Apache POI**

- Excel Data Reading
- Data Driven Testing

**Alerts Handling**

- Simple Alert
- Confirmation Alert
- Prompt Alert

**Window Handling**

- Multiple Windows
- Multiple Tabs

**Actions Class**

- Mouse Hover
- Double Click
- Drag and Drop
- Slider Handling

**Web Tables**

- Row Validation
- Data Verification

**File Upload**

- Single File Upload
- Multiple File Upload

**Other Modules**

- Dropdown Handling
- Date Picker
- Broken Links
- Search Functionality

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**Automated Test Cases:**

**Form Validation**

1. Verify Name Field
2. Verify Email Field
3. Verify Phone Field
4. Verify Address Field

**Radio Buttons**

5. Verify Male Radio Button

**Checkboxes**

6. Verify All Day Checkboxes

**Dropdowns**

7. Verify Country Dropdown
8. Verify Color Dropdown
9. Verify Animal Dropdown

**Date Picker**

10. Verify Date Selection

**File Upload**

11. Verify Single File Upload
12. Verify Multiple File Upload

**Search Functionality**

13. Verify Search

**Alert Handling**

14. Verify Simple Alert
15. Verify Confirm Alert
16. Verify Prompt Alert

**Window Handling**

17. Verify New Tab
18. Verify Popup Window

**Actions Class**

19. Verify Mouse Hover
20. Verify Double Click
21. Verify Drag and Drop
22. Verify Slider

**Web Tables**

23. Verify Table Rows

**Links**

24. Verify Broken Links

## General

25. Verify Page Title

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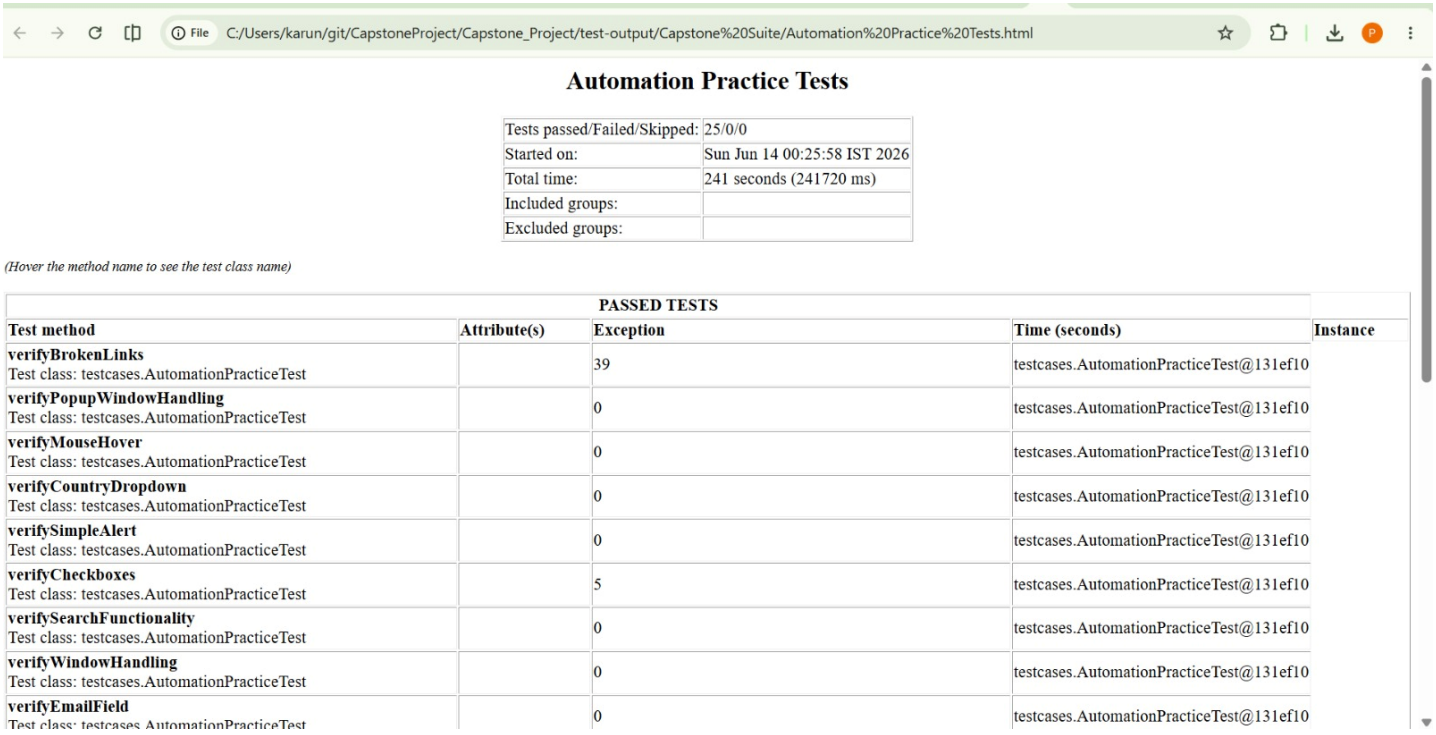
## Automation Test Report:

After successful execution of the automation suite, TestNG generates a detailed execution report that provides information about the overall test execution status. The report shown above demonstrates that all automated test cases executed successfully without any failures or skipped tests.



File C:/Users/karun/git/CapstoneProject/Capstone_Project/test-output/Capstone%20Suite/Automation%20Practice%20Tests.html

**Automation Practice Tests**

| | |
|---|---|
| Tests passed/Failed/Skipped: | 25/0/0 |
| Started on: | Sun Jun 14 00:25:58 IST 2026 |
| Total time: | 241 seconds (241720 ms) |
| Included groups: | |
| Excluded groups: | |

*(Hover the method name to see the test class name)*

**PASSED TESTS**

| Test method | Attribute(s) | Exception | Time (seconds) | Instance |
|---|---|---|---|---|
| **verifyBrokenLinks** Test class: testcases.AutomationPracticeTest | | 39 | testcases.AutomationPracticeTest@131ef10 | |
| **verifyPopupWindowHandling** Test class: testcases.AutomationPracticeTest | | 0 | testcases.AutomationPracticeTest@131ef10 | |
| **verifyMouseHover** Test class: testcases.AutomationPracticeTest | | 0 | testcases.AutomationPracticeTest@131ef10 | |
| **verifyCountryDropdown** Test class: testcases.AutomationPracticeTest | | 0 | testcases.AutomationPracticeTest@131ef10 | |
| **verifySimpleAlert** Test class: testcases.AutomationPracticeTest | | 0 | testcases.AutomationPracticeTest@131ef10 | |
| **verifyCheckboxes** Test class: testcases.AutomationPracticeTest | | 5 | testcases.AutomationPracticeTest@131ef10 | |
| **verifySearchFunctionality** Test class: testcases.AutomationPracticeTest | | 0 | testcases.AutomationPracticeTest@131ef10 | |
| **verifyWindowHandling** Test class: testcases.AutomationPracticeTest | | 0 | testcases.AutomationPracticeTest@131ef10 | |
| **verifyEmailField** Test class: testcases.AutomationPracticeTest | | 0 | testcases.AutomationPracticeTest@131ef10 | |

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**Defects Identified:**

The following validation defects were observed:

- Name field accepts numeric values.
- Email field accepts invalid email format.
- Mobile number accepts less than 10 digits.
- Address accepts special characters only.
- Broken Link Returns 404 Error

These defects indicate missing input validation in the application. This aligns with the defect analysis documented in the uploaded report.

**Defect Report:**

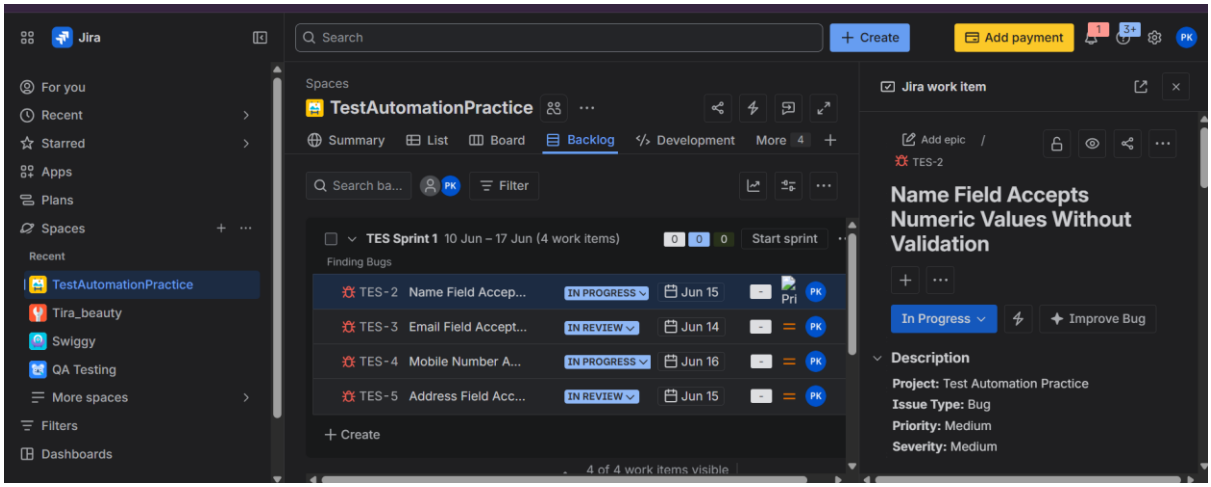
**BUG-01:** Name Field Accepts Numeric Values Without Validation

**Steps to Reproduce:**

1. Open the application.
2. Enter 123456 in the Name field.
3. Submit the form.

**Expected Result:**
The Name field should accept only alphabetic characters.

**Actual Result:**
Numeric values are accepted without any validation message



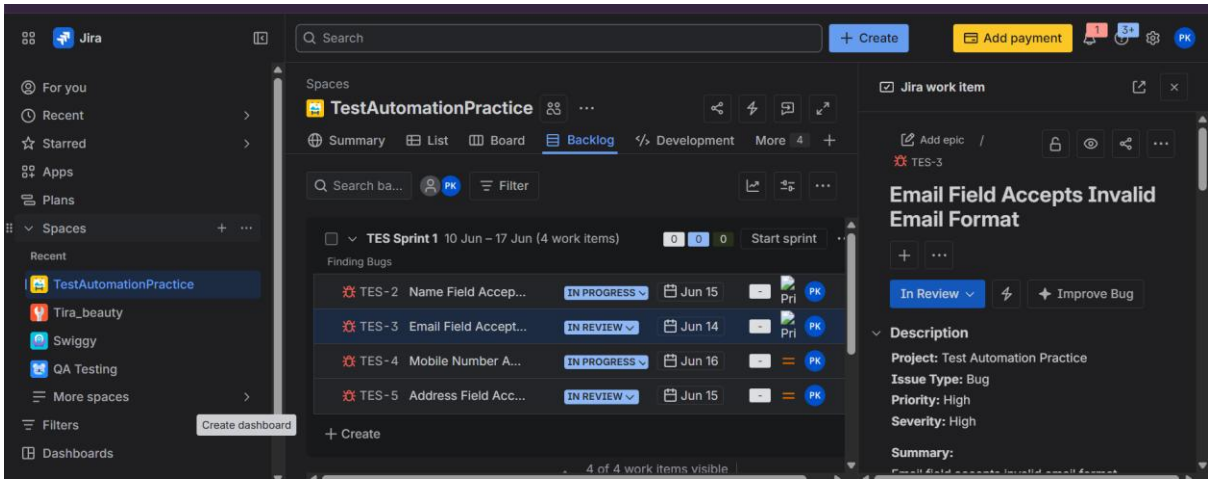

**BUG-02**: Email Field Accepts Invalid Email Format

**Steps to Reproduce:**

1. Open the application.
2. Enter karunakar123 in the Email field.
3. Submit the form.

**Expected Result:**
The Email field should accept only valid email formats containing @ and domain.

**Actual Result:**
Invalid email is accepted without validation.



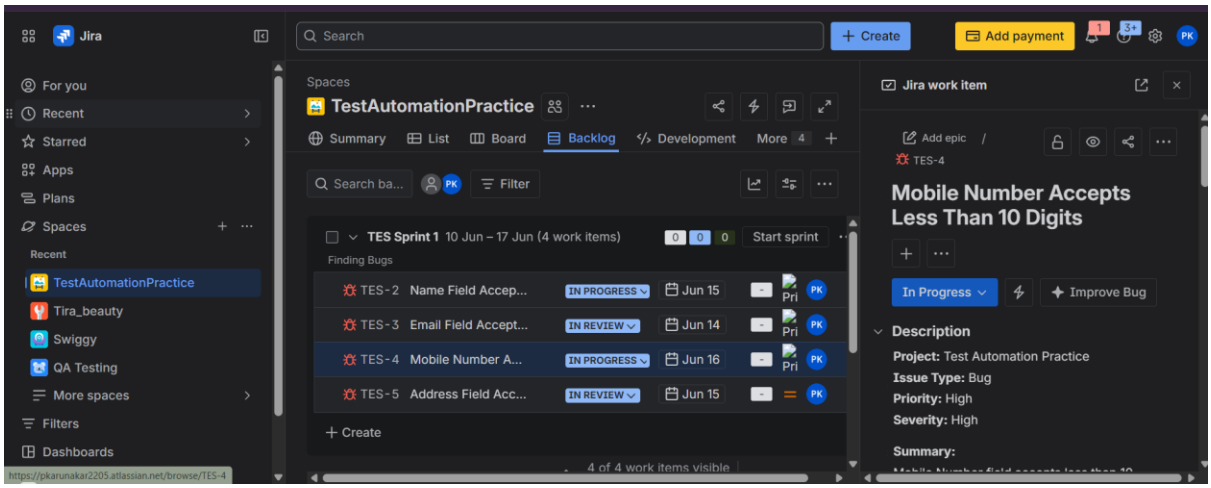
**BUG-03**: Mobile Number Accepts Less Than 10 Digits

**Steps to Reproduce:**

1. Open the application.
2. Enter 12345 in the Mobile Number field.
3. Submit the form.

**Expected Result:**
The Mobile Number field should accept exactly 10 digits.

**Actual Result:**
Application accepts less than 10 digits without validation.



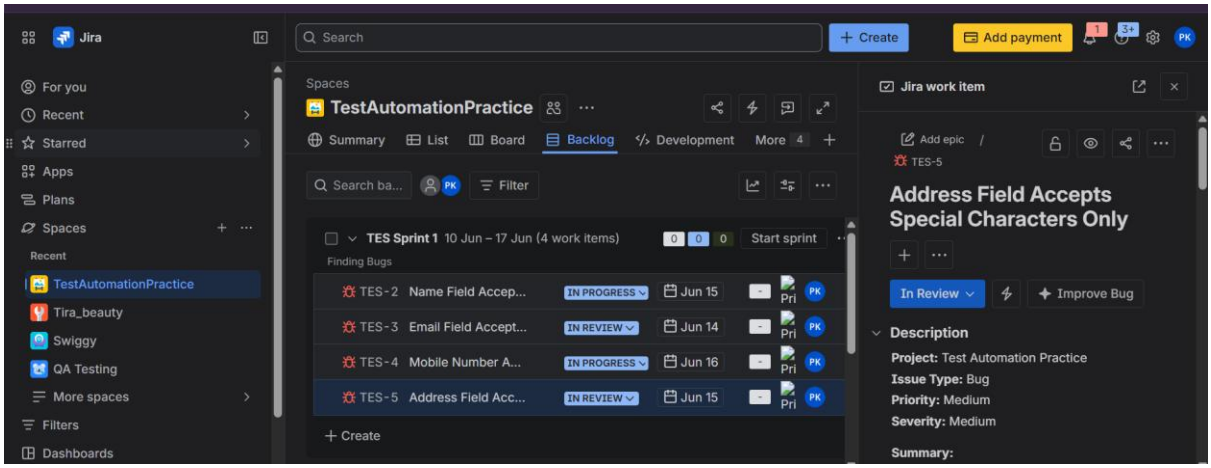

**BUG-04:** Address Field Accepts Special Characters Only

**Steps to Reproduce:**

1. Open the application.
2. Enter @#$%^&*() in the Address field.
3. Submit the form.

**Expected Result:**
Address field should contain meaningful address information.

**Actual Result:**
Special characters are accepted without validation.



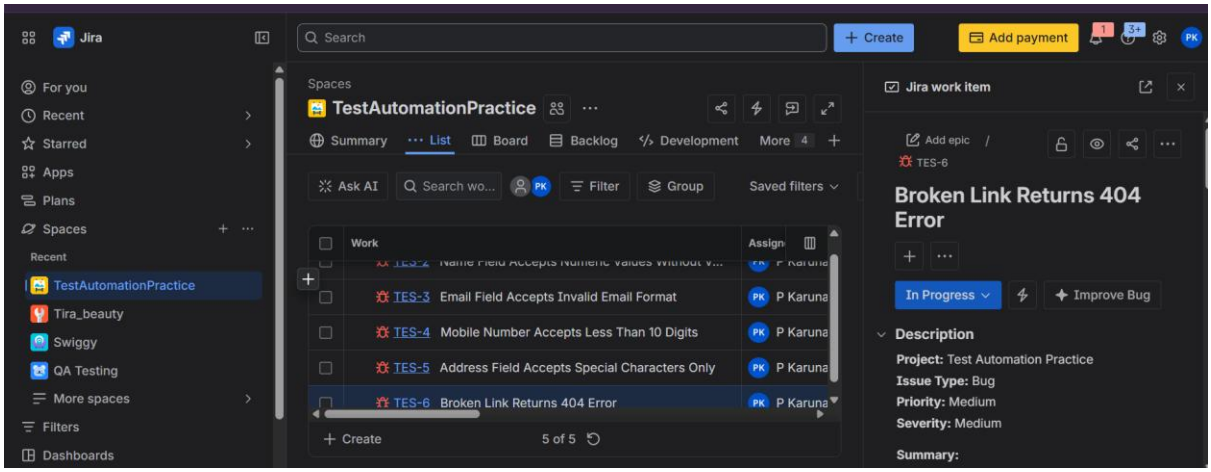

**BUG-05:** Broken Link Returns 404 Error

**Steps to Reproduce:**

1. Navigate to the application.
2. Click available links or run broken link verification.
3. Observe response codes.

**Expected Result:**
All links should return valid HTTP response codes (200 OK).

**Actual Result:**
Some links return HTTP 404 Not Found.




## Defect Summary Table:

| Bug ID | Summary | Severity | Priority |
|---|---|---|---|
| BUG-01 | Name field accepts numeric values | Medium | Medium |
| BUG-02 | Invalid email accepted | High | High |
| BUG-03 | Mobile number less than 10 digits accepted | High | High |
| BUG-04 | Address field accepts special characters | Medium | Medium |
| BUG-05 | Broken link returns 404 | Medium | Medium |

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**Jenkins Integration:**

Jenkins was integrated with GitHub to achieve Continuous Integration.

Execution Flow:

GitHub Repository

↓

Jenkins Pipeline

↓

Maven Build

↓

TestNG Execution

↓

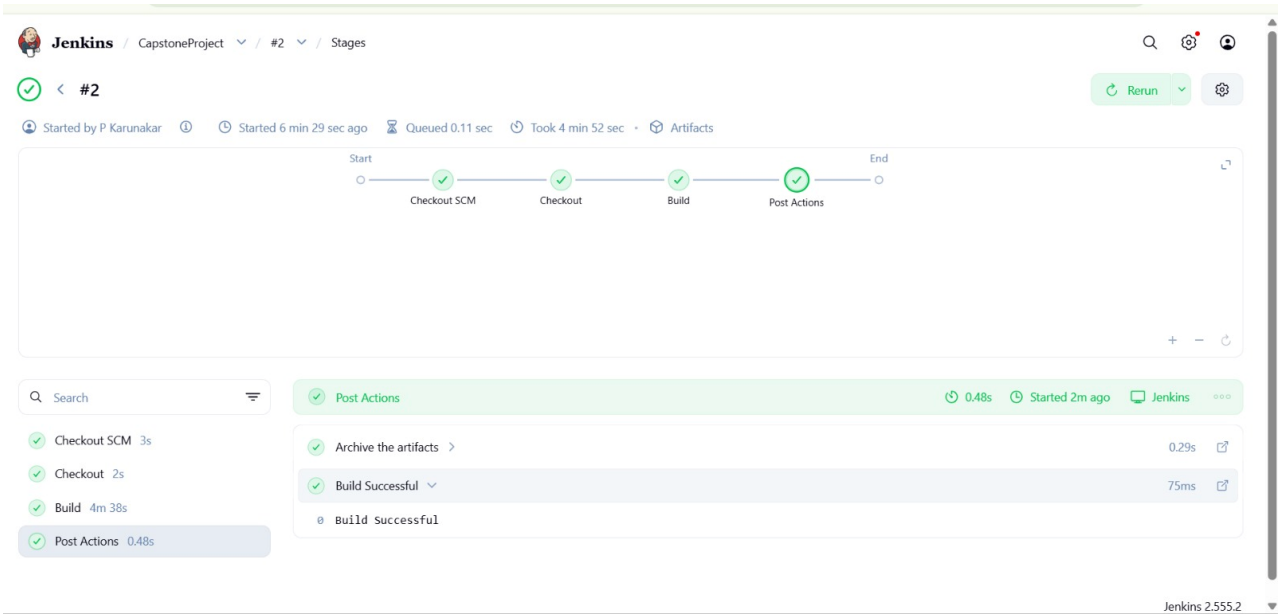
Report Generation

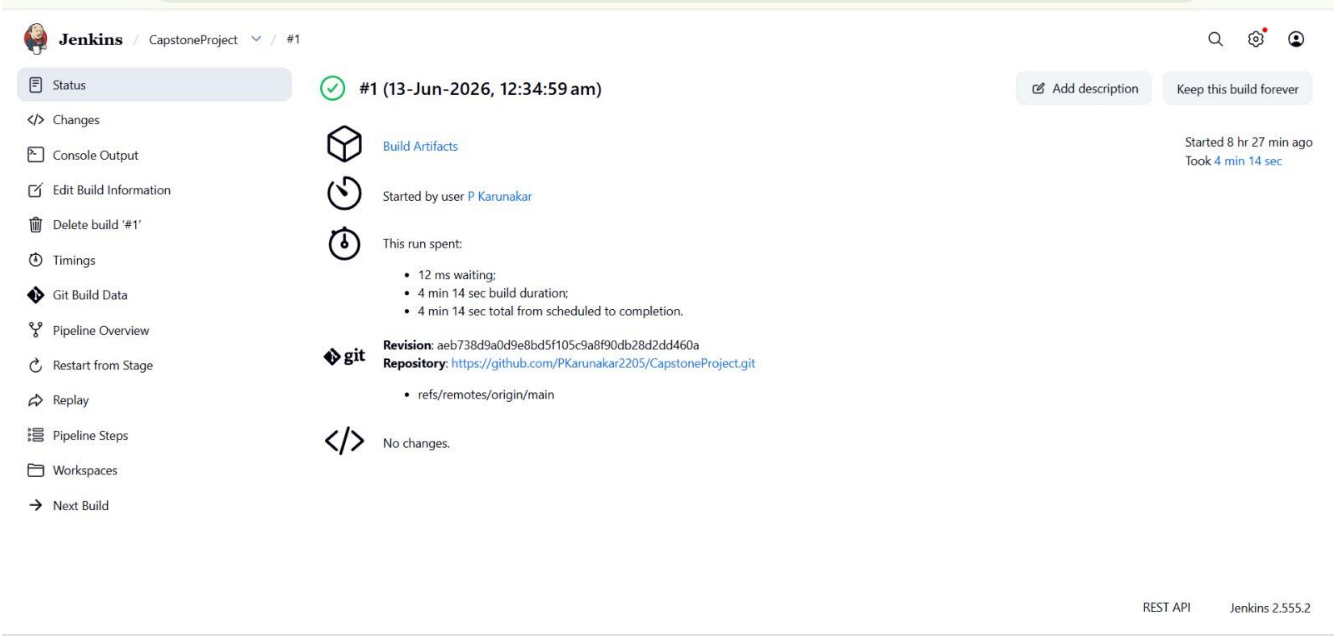
↓

Build Status

Benefits:

- Automated execution
- Continuous testing
- Faster feedback
- Easy monitoring
- Build history maintenance


Jenkins / CapstoneProject / #2 / Stages
#2
Rerun
Started by P Karunakar
Started 6 min 29 sec ago
Queued 0.11 sec
Took 4 min 52 sec
Artifacts
Start
End
Checkout SCM
Checkout
Build
Post Actions
Search
Checkout SCM 3s
Checkout 2s
Build 4m 38s
Post Actions 0.48s
Post Actions
0.48s
Started 2m ago
Jenkins
Archive the artifacts
0.29s
Build Successful
75ms
Build Successful
Jenkins 2.555.2


Jenkins / CapstoneProject / #1
Status
Changes
Console Output
Edit Build Information
Delete build '#1'
Timings
Git Build Data
Pipeline Overview
Restart from Stage
Replay
Pipeline Steps
Workspaces
Next Build
#1 (13-Jun-2026, 12:34:59 am)
Add description
Keep this build forever
Build Artifacts
Started 8 hr 27 min ago
Took 4 min 14 sec
Started by user P Karunakar
This run spent:
12 ms waiting;
4 min 14 sec build duration;
4 min 14 sec total from scheduled to completion.
Revision: aeb738d9a0d9e8bd5f105c9a8f90db28d2dd460a
Repository: https://github.com/PKarunakar2205/CapstoneProject.git
refs/remotes/origin/main
No changes.
REST API
Jenkins 2.555.2

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**GitHub Repository:**

Repository URL:

https://github.com/PKarunakar2205/CapstoneProject

GitHub is used for:

- Source Code Management
- Version Control
- Collaboration
- Jenkins Integration

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**Test Execution Summary:**

| Metric | Count |
|---|---|
| Total Test Cases | 25 |
| Executed Test Cases | 25 |
| Passed Test Cases | 25 |
| Failed Test Cases | 0 |
| Blocked Test Cases | 0 |

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**Challenges Faced:**

- Jenkins configuration and setup
- Maven build issues
- File path handling
- Apache POI Excel integration
- Selenium locator maintenance
- Browser compatibility issues
- GitHub integration
- Dynamic element synchronization

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**Conclusion:**

The Automation Testing Framework successfully automated the major functionalities of the Test Automation Practice Website using Selenium WebDriver, Java, TestNG, Maven, Jenkins, GitHub, and Apache POI.

The framework follows industry-standard Page Object Model architecture and provides reusable, maintainable, and scalable automation components. Continuous Integration was achieved using Jenkins, and version control was managed through GitHub.

The framework improved execution efficiency, reduced manual effort, increased test coverage, and enabled reliable regression testing. All planned automation scenarios were successfully implemented and executed.